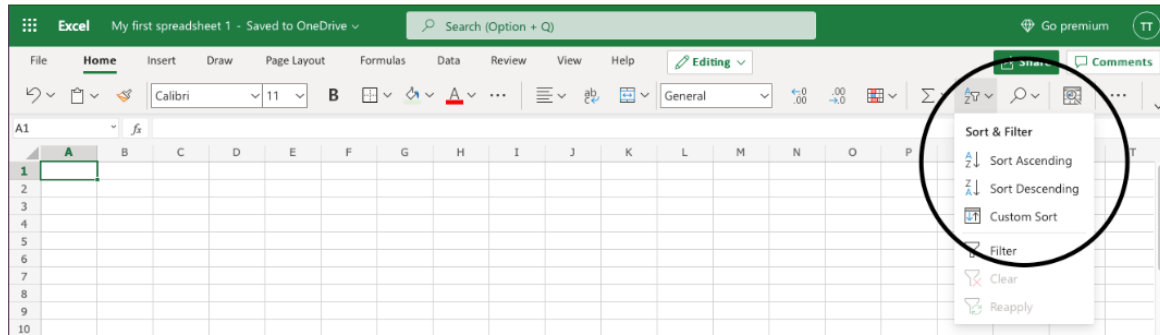


Assignment-2-CT- Group Work 21-12-2025 → done

Activity:

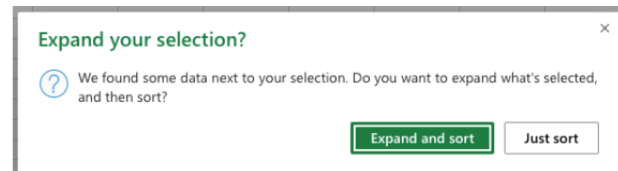
Excel for Data Analysis

1- Excel Sorting

Ranges can be sorted using the **Sort Ascending** and **Sort Descending** commands.**A Non-Working Example (sorting one column in a range)**

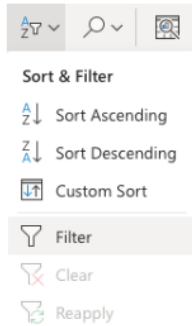
In this example we have two columns with related data. Column **A** is the Pokemons **Names** and Column **B** is their **Total stats**. Try sorting just one of the columns (**A2:A21**) ascending by their **Names**.

	A	B	C
1	Name	Total	
2	Bulbasaur	318	
3	Ivysaur	405	
4	Venusaur	525	
5	Charmander	309	
6	Charmeleon	405	
7	Charizard	534	
8	Squirtle	314	
9	Wartortle	405	
10	Blastoise	530	
11	Caterpie	195	
12	Metapod	205	
13	Butterfree	395	
14	Weedle	195	
15	Kakuna	205	
16	Beedrill	395	
17	Pidgey	251	
18	Pidgeotto	349	
19	Pidgeot	479	
20	Rattata	253	
21	Raticate	413	



2- Excel Filter

The menu is accessed in the default Ribbon view or in the **Data** section in the navigation bar.



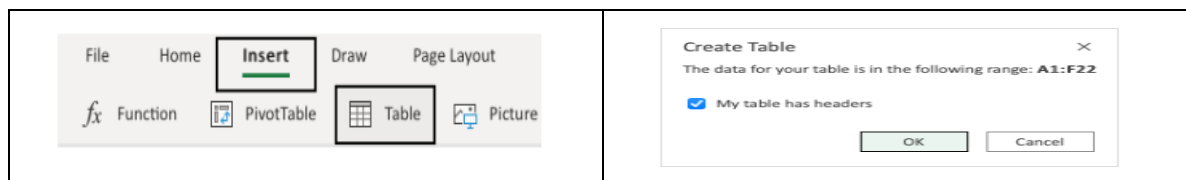
3- Excel Tables

Example: Formatting a range into a table will give it a new form with a fixed structure. Tables open access to new functionality such as: filtering, automations and styling.

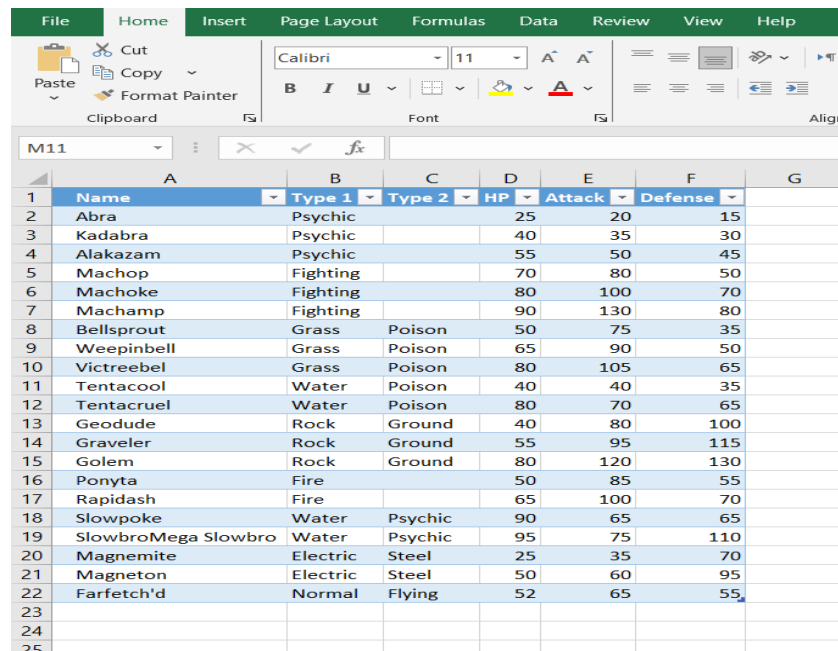
	A	B	C	D	E	F
1	Name	Type 1	Type 2	HP	Attack	Defense
2	Abra	Psychic		25	20	15
3	Kadabra	Psychic		40	35	30
4	Alakazam	Psychic		55	50	45
5	Machop	Fighting		70	80	50
6	Machoke	Fighting		80	100	70
7	Machamp	Fighting		90	130	80
8	Bellsprout	Grass	Poison	50	75	35
9	Weepinbell	Grass	Poison	65	90	50
10	Victreebel	Grass	Poison	80	105	65
11	Tentacool	Water	Poison	40	40	35
12	Tentacruel	Water	Poison	80	70	65
13	Geodude	Rock	Ground	40	80	100
14	Graveler	Rock	Ground	55	95	115
15	Golem	Rock	Ground	80	120	130
16	Ponyta	Fire		50	85	55
17	Rapidash	Fire		65	100	70
18	Slowpoke	Water	Psychic	90	65	65
19	Slowbro	Water	Psychic	95	75	110
20	Magnemite	Electric	Steel	25	35	70
21	Magneton	Electric	Steel	50	60	95
22	Farfetch'd	Normal	Flying	52	65	55

3.1. Select range **A1:F22**

3.2. Click **Insert**, then Table ( Table), in the Ribbon.



3.3. Click OK



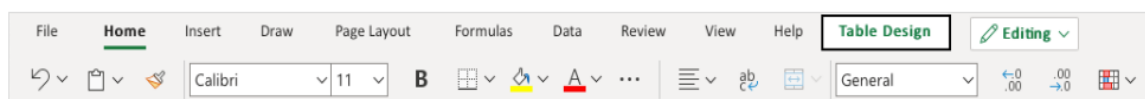
	A	B	C	D	E	F	G
	Name	Type 1	Type 2	HP	Attack	Defense	
2	Abra	Psychic		25	20	15	
3	Kadabra	Psychic		40	35	30	
4	Alakazam	Psychic		55	50	45	
5	Machop	Fighting		70	80	50	
6	Machoke	Fighting		80	100	70	
7	Machop	Fighting		90	130	80	
8	Bellsprout	Grass	Poison	50	75	35	
9	Weepinbell	Grass	Poison	65	90	50	
10	Victreebel	Grass	Poison	80	105	65	
11	Tentacool	Water	Poison	40	40	35	
12	Tentacool	Water	Poison	80	70	65	
13	Geodude	Rock	Ground	40	80	100	
14	Graveler	Rock	Ground	55	95	115	
15	Golem	Rock	Ground	80	120	130	
16	Ponyta	Fire		50	85	55	
17	Rapidash	Fire		65	100	70	
18	Slowpoke	Water	Psychic	90	65	65	
19	Slowbro	Water	Psychic	95	75	110	
20	Magnemite	Electric	Steel	25	35	70	
21	Magnetron	Electric	Steel	50	60	95	
22	Farfetch'd	Normal	Flying	52	65	55	
23							
24							
25							

Table Design

Tables can be customized and styled in a few clicks. Converting a range into a table gives access to a menu called "Table Design".

This menu has options and commands such as:

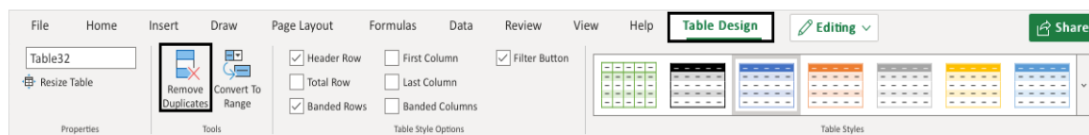
- Resize
- Remove duplicates
- Convert to range
- Style options (Total row, Header row, Banded row etc.)
- Formatting



Let's resize the table from range **A1:F12** to **A1:F20**

	A	B	C	D	E	F	G
1	Name	Type 1	Type 2	HP	Attack	Defense	
2	Abra	Psychic		25	20	15	
3	Kadabra	Psychic		40	35	30	
4	Alakazam	Psychic		55	50	45	
5	Machop	Fighting		70	80	50	
6	Machoke	Fighting		80	100	70	
7	Machamp	Fighting		90	130	80	
8	Bellsprout	Grass	Poison	50	75	35	
9	Weepinbell	Grass	Poison	65	90	50	
10	Victreebel	Grass	Poison	80	105	65	
11	Tentacool	Water	Poison	40	40	35	
12	Tentacruel	Water	Poison	80	70	65	
13							

Removing Duplicates



Excel Table Style as well ...

4- Excel Conditional Formatting

Conditional formatting is used to change the appearance of cells in a range based on your specified conditions. The conditions are rules based on specified numerical values or matching text.

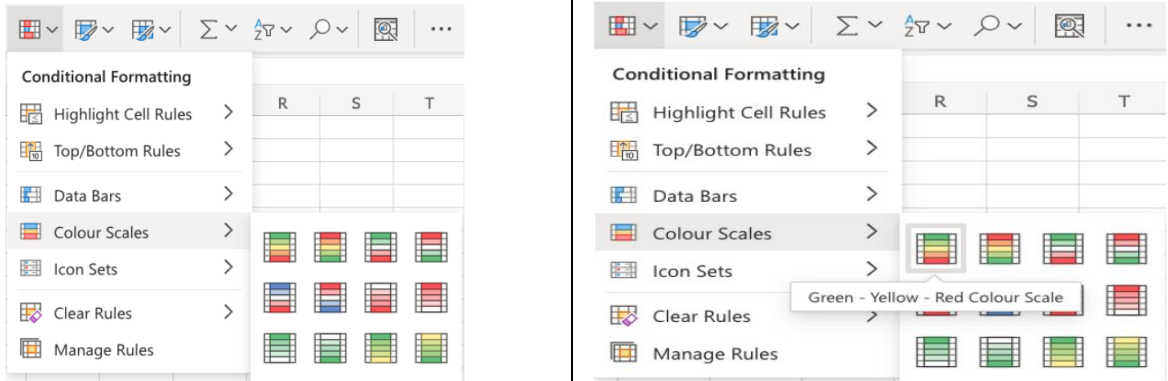
--

each pokemon is formatted with a **Color Scale**:

	A	B	C	D
1	Name	Type 1	Speed	
2	Bulbasaur	Grass	45	
3	Ivysaur	Grass	60	
4	Venusaur	Grass	80	
5	Charmander	Fire	65	
6	Charmeleon	Fire	80	
7	Charizard	Fire	100	
8	Squirtle	Water	43	
9	Wartortle	Water	58	
10				

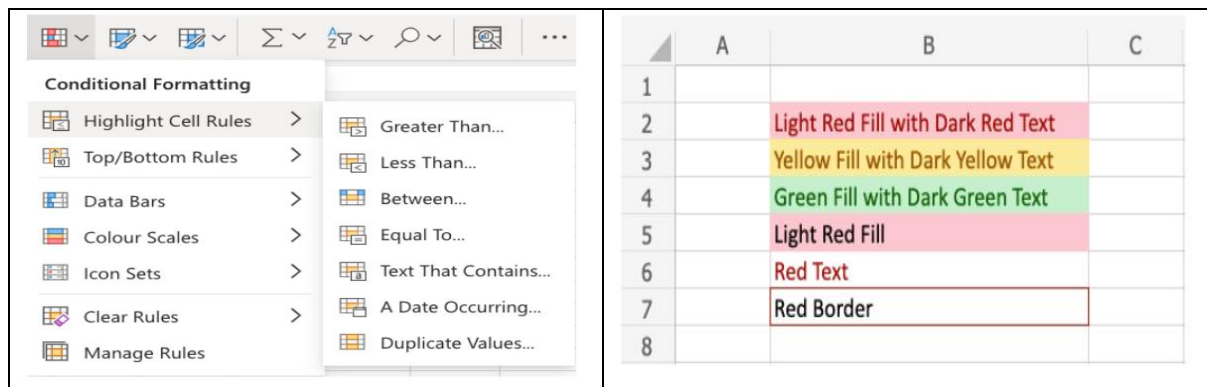
Highlight the Speed values of each Pokemon with Color scale conditional formatting.

- 1- Select the range of Speed values **C2:C9**
- 2- Click on the Conditional Formatting icon  in the ribbon, from the **Home** menu
- 3- Select **Color Scales** from the drop-down menu



Highlight Cell Rules:

Highlight Cell Rules is a premade type of conditional formatting in Excel used to change the appearance of cells in a range based on your specified conditions. The conditions are rules based on specified numerical values, matching text, calendar dates, or duplicated and unique values.



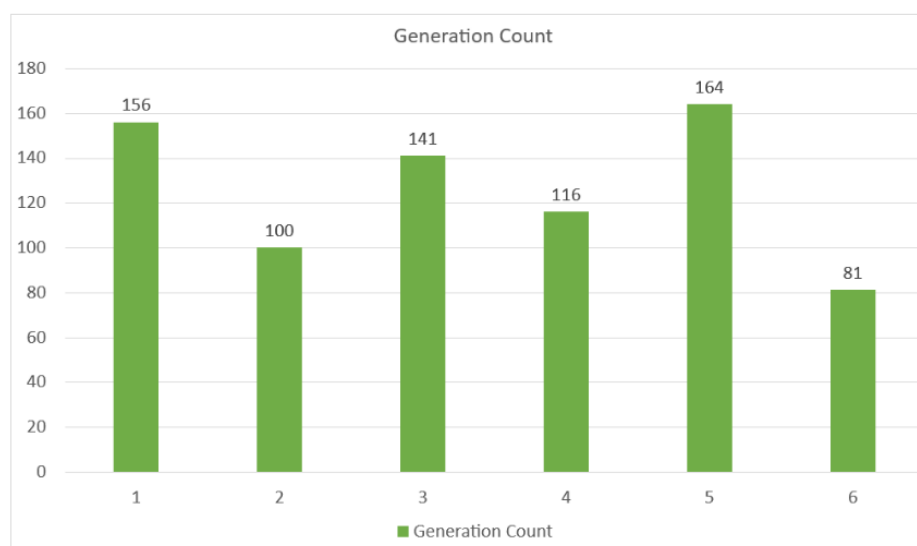
Appearance Options: The web browser version of Excel offers the following appearance options for conditionally formatted cells:

- Light Red Fill with Dark Red Text
- Yellow Fill with Dark Yellow Text
- Green Fill with Dark Green Text
- Light Red Fill
- Red Text
- Red Border

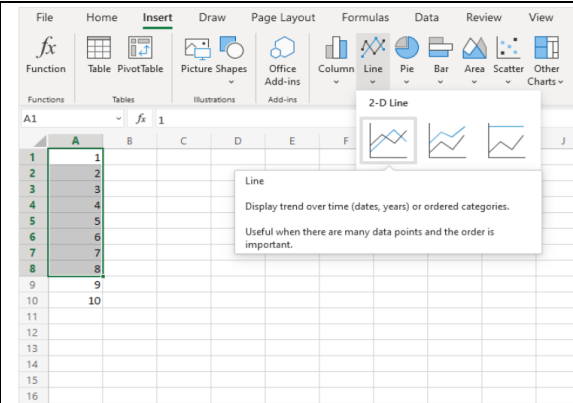
Excel Charts:

Charts are visual representations of data used to make it more understandable. Commonly used charts are:

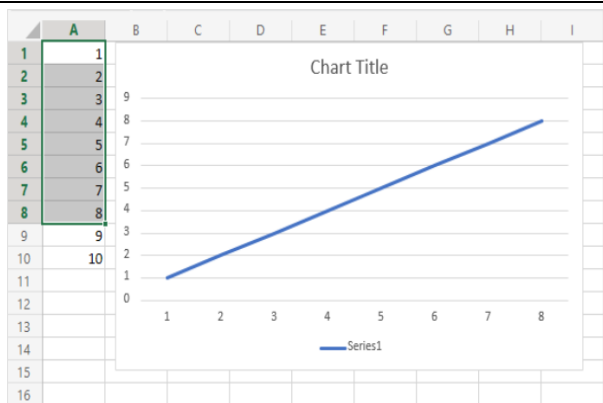
- Pie chart
- Column chart
- Line chart



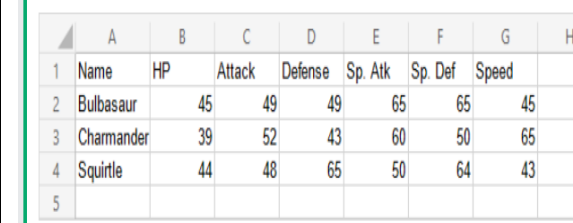
2. Click on the **Insert** menu, then click on the Line menu () and choose **Line** () from the drop-down menu



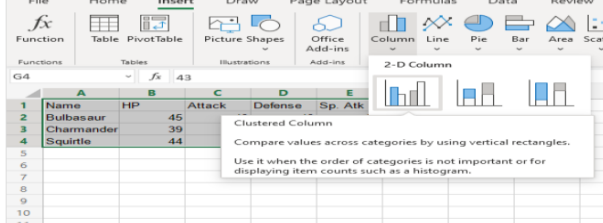
The screenshot shows the Excel ribbon with the **Insert** tab selected. The **Line** chart icon is highlighted in the **Charts** group. A tooltip for the **Line** chart is displayed, stating: "Line. Display trend over time (dates, years) or ordered categories. Useful when there are many data points and the order is important."



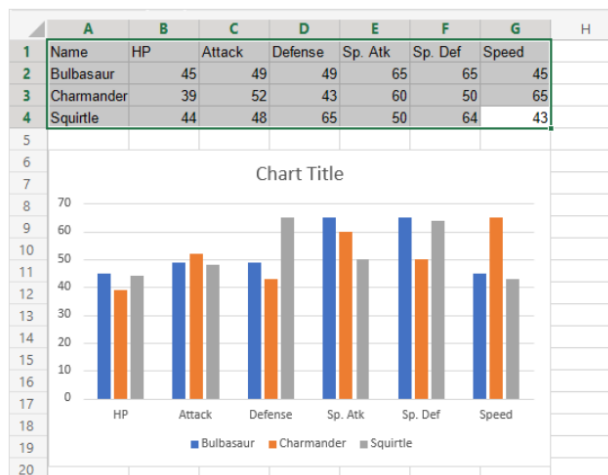
The screenshot shows a line chart titled "Chart Title" plotted on a grid. The x-axis is labeled 1 through 8, and the y-axis is labeled 0 through 10. A single blue line, labeled "Series1", starts at (1, 1) and ends at (8, 8), showing a linear upward trend.



	A	B	C	D	E	F	G	H
1	Name	HP	Attack	Defense	Sp. Atk	Sp. Def	Speed	
2	Bulbasaur	45	49	49	65	65	45	
3	Charmander	39	52	43	60	50	65	
4	Squirtle	44	48	65	50	64	43	
5								



The screenshot shows the Excel ribbon with the **Insert** tab selected. The **Column** chart icon is highlighted in the **Charts** group. A tooltip for the **Clustered Column** chart is displayed, stating: "Clustered Column. Compare values across categories by using vertical rectangles. Use it when the order of categories is not important or for displaying item counts such as a histogram."



5- Excel Functions

Before we apply functions let us see truth table (Boolean Algebra) ¹.

<table><tr><th colspan="3">Truth Table</th></tr><tr><td>A</td><td>B</td><td>Q</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td colspan="3">Read as A AND B gives Q</td></tr></table>	Truth Table			A	B	Q	0	0	0	0	1	0	1	0	0	1	1	1	Read as A AND B gives Q			<table><tr><th colspan="3">Truth Table</th></tr><tr><td>A</td><td>B</td><td>Q</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td colspan="3">Read as A OR B gives Q</td></tr></table>	Truth Table			A	B	Q	0	0	0	0	1	1	1	0	1	1	1	1	Read as A OR B gives Q		
Truth Table																																											
A	B	Q																																									
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0	1	1																																									
1	0	1																																									
1	1	1																																									
Read as A OR B gives Q																																											
<table><tr><th colspan="2">Truth Table</th></tr><tr><td>A</td><td>Q</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr><tr><td colspan="2">Read as inversion of A gives Q</td></tr></table>	Truth Table		A	Q	0	1	1	0	Read as inversion of A gives Q		Question: what if we have 3 Inputs ? How can we represent a truth table ?																																
Truth Table																																											
A	Q																																										
0	1																																										
1	0																																										
Read as inversion of A gives Q																																											

** Calculated as $2^{(\text{number of Input(s)})}$

AND Function: returns **TRUE** or **FALSE** based on two or more **conditions**.

=AND([logical1], [logical2], ...) The conditions can check things like:

- If a number is **greater than** another number **>**
- If a number is **smaller than** another number **<**
- If a number or text is **equal** to something **=**

Example **AND** function, step by step:

1. Select the cell D2
2. Type =AND
3. Double click the **AND** command

¹ https://www.electronics-tutorials.ws/boolean/bool_7.html

4. Specify the first condition $B2="Fire"$
5. Type ,
6. Specify the second condition $C2>70$
7. Hit enter

	A	B	C	D	E
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed	
2	Bulbasaur	Grass	45		
3	Ivysaur	Grass	60		
4	Venusaur	Grass	80		
5	Charmander	Fire	65		
6	Charmeleon	Fire	80		
7	Charizard	Fire	100		
8	Squirtle	Water	43		
9	Wartortle	Water	58		
10	Blastoise	Water	78		
11					

	A	B	C	D	E	F	G	H	I
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed					
2	Bulbasaur	Grass	45	=AND					
3	Ivysaur	Grass	60	AND					
4	Venusaur	Grass	80	BITAND					
5	Charmander	Fire	65	RAND					
6	Charmeleon	Fire	80	RANDARRAY					
7	Charizard	Fire	100	RANDBETWEEN					
8	Squirtle	Water	43	STANDARDIZE					
9	Wartortle	Water	58						
10	Blastoise	Water	78						
11									

	A	B	C	D	E
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed	
2	Bulbasaur	Grass	45	=AND(B2="Fire"; C2>70)	
3	Ivysaur	Grass	60	AND (logical1; [logical2]; [logical3]; ...)	
4	Venusaur	Grass	80		
5	Charmander	Fire	65		
6	Charmeleon	Fire	80		
7	Charizard	Fire	100		
8	Squirtle	Water	43		
9	Wartortle	Water	58		
10	Blastoise	Water	78		
11					

	A	B	C	D	E
1	Name	Type 1	Speed	Fire Type AND more than 70 Speed	
2	Bulbasaur	Grass	45	=AND(B2="Fire"; C2>70)	
3	Ivysaur	Grass	60	=AND(B3="Fire"; C3>70)	
4	Venusaur	Grass	80	=AND(B4="Fire"; C4>70)	
5	Charmander	Fire	65	=AND(B5="Fire"; C5>70)	
6	Charmeleon	Fire	80	=AND(B6="Fire"; C6>70)	
7	Charizard	Fire	100	=AND(B7="Fire"; C7>70)	
8	Squirtle	Water	43	=AND(B8="Fire"; C8>70)	
9	Wartortle	Water	58	=AND(B9="Fire"; C9>70)	
10	Blastoise	Water	78	=AND(B10="Fire"; C10>70)	
11					

AVERAGE Function:

The AVERAGE function is a premade function in Excel, which calculates the average (arithmetic mean). It is typed **=AVERAGE** It adds the range and divides it by the number of observations.

F2	A	B	C	D	E	F
1	Trainer	Pokeball	Great ball	Ultraball	Master ball	Average
2	Iva	10	4	1	1	
3	Liam	12	3	0	1	
4	Jenny	15	1	3	1	
5	Iben	4	2	6	0	
6	Adora	10	4	1	1	
7	Kasper	9	2	1	0	

F2	A	B	C	D	E	F	G	H
1	Trainer	Pokeball	Great ball	Ultraball	Master ball	Average		
2	Iva	10	4	1	1	=AVERAGE		
3	Liam	12	3	0	1	AVERAGE (number1; [number2]; ...)		
4	Jenny	15	1	3	1			
5	Iben	4	2	6	0			
6	Adora	10	4	1	1			
7	Kasper	9	2	1	0			

COUNT Function

The **COUNT** function is a premade function in Excel, which counts cells with numbers in a range.

It is typed =COUNT

Note: The **COUNT** function only counts cells with numbers, **not** cells **with letters**.

How to use the =COUNT function:

1. Select a cell
2. Type =COUNT
3. Double click the **COUNT** command
4. Select a range
5. Hit enter